

UMAR HAMZAH

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PROFILE SUMMARY

First-batch Information Systems student at Universitas Cakrawala with a strong technical foundation in Software Engineering from SMK Telkom Jakarta. Experienced in IoT development, robotics education, and full-stack software development across web and mobile platforms. Participated in Gemastik XVIII 2025 (Smart Devices category), developing a machine learning-powered pilgrim coordination system. Skilled in mobile development (Flutter), web development (React, Node.js, Express), database ORM (Prisma), embedded systems, and AI-based anomaly detection. A collaborative team player with strong problem-solving skills and a passion for engineering impactful solutions.

EDUCATION

B.Sc. Information Systems — Universitas Cakrawala (Founding Batch) 2024 – Present

- Gemastik XVIII 2025 Finalist — national competition in Smart Devices category, organized by Ministry of Education x Universitas Telkom.
- Actively building cross-subject projects spanning mobile development, full-stack web, IoT, and AI domains.

Vocational High School — Software Engineering — SMK Telkom Jakarta July 2021 – June 2024

INTERNSHIP EXPERIENCE

IoT Developer Intern | PT Global Inovasi Terdepan (RightJet) 2025

- Developed an internet-free offline communication system using ESP32 as a coordinator device for Umrah and Hajj pilgrims.
- Designed an ESP32-based primary channel connecting directly to pilgrims' smartphones, replacing conventional communication tools in hardware-constrained environments.
- Applied embedded systems and wireless communication concepts to optimize signal transmission reliability.

Assistant Teacher — Robotics Education | PT Sari Teknologi July 2023 – December 2023

- Assisted robotics learning sessions across 4 schools, from elementary to junior high school levels.
- Designed and prepared customized teaching materials tailored to student competency assessments.
- Developed IoT competency through a collaborative group project focused on IoT-based smart automation systems.

PROJECT EXPERIENCE

GOALPOST — Community-Based Football Portal & Match Tracker Semester 3, 2025

Web Application Development 2 (UAS) | Stack: React (Vite), Node.js, Express, Prisma ORM, PostgreSQL, JWT Authentication, CSS

- Developed a full-stack community-driven football portal and article hub for football enthusiasts.
- Built interactive community features: publish articles, upload multiple images, comment, like, and save bookmarks.
- Implemented multi-role access control (Admin / User) with JWT authentication and secure password hashing.
- Designed relational database schema in Prisma ORM with composite primary keys for Likes and Bookmarks, and cascade deletion rules.
- Engineered a match-tracking and scheduler dashboard to record tournament phases — World Cup scores, cards, and team statistics.

Aplikasi Buku Masak (Recipe App) — Multi-User Mobile App Semester 3, 2025

Mobile Computing (UAS) | Stack: Flutter (Dart), Provider, Hive (NoSQL), TheMealDB API, Image Picker

- Built a multi-user recipe catalog mobile app with data isolation per user account and persistent auto-login session.
- Integrated TheMealDB API for real-time online recipe search and local import functionality.
- Developed custom recipe creation with device camera integration and local Base64 image storage.
- Built key features: real-time search, favorite list, smart shopping list, and Admin Mode for central data management.

Sistem Koordinasi Jamaah Adaptif — Gemastik XVIII 2025 (Smart Devices)

October 2025

Universitas Cakrawala — National Competition | Stack: ESP32-WROOM-32, GPS NEO-8M, ESP-NOW Mesh, DBSCAN, One-Class SVM, MapLibre, Arduino IDE

- Designed a hybrid dual-mode IoT platform for pilgrim coordination — offline via ESP-NOW mesh and online via cloud sync.
- Implemented two-layer anomaly detection (DBSCAN + One-Class SVM) achieving 94% accuracy and only 6% false alarm rate.
- Built key features: offline real-time map, SOS emergency trigger, Push-to-Talk voice, and digital Tawaf & Sai counter.

TrakPoint — Smart Fleet Monitoring System

Semester 2, 2024

Web Client Development | UX Design | Database | Stack: PHP, HTML, CSS, MySQL, Node.js, WebSocket, Mosquitto (MQTT), ESP-32, GPS NEO-6M, Figma

- Developed a web-based truck fleet monitoring system with real-time GPS tracking, overspeed alarms, idle detection, and trip history.
- Managed database design (MySQL) and IoT hardware integration (ESP-32, GPS NEO-6M, SIM800L) for location tracking.
- Designed UI/UX in Figma and implemented a multi-role user system (Admin / Operator / Driver).

TransportKu — Smart City Transportation Assistant

Semester 3, 2025

Information Systems Concept | Stack: Figma, Use Case Diagram, BPMN, Activity Diagram

- Conceptualized an integrated city transportation platform combining train/bus ticketing and ride-hailing services.
- Produced full system design documentation and designed 30+ responsive UI/UX screens in Figma.

TECHNICAL SKILLS

Programming Languages: HTML, CSS, PHP, JavaScript (ES6+), Dart, C++, SQL

Frameworks & Libraries: Flutter, React, Node.js, Express, Prisma ORM, JWT

Databases: MySQL, PostgreSQL, Hive (NoSQL), SQLite

Hardware & IoT: ESP-32, ESP-NOW Mesh, GPS NEO-6M/NEO-8M, SIM800L, RFID, Arduino IDE

Tools & Software: Git, GitHub, Figma, VS Code, Mosquitto (MQTT), WebSocket, MapLibre, Tableau, Canva

CERTIFICATIONS

Certificate of Appreciation — IoT Developer Intern | PT Global Inovasi Terdepan (RightJet) | November 2025

Certificate of Participation — Gemastik XVIII (Smart Devices) | Ministry of Education x Universitas Telkom | October 2025

Certificate of Competence — Junior Coder | BNSP (Indonesian Professional Certification Authority) | May 2024

Vocational Competency Certificate — Web School Complaint App | SMK Telkom Jakarta | Score: 89/100 | June 2024

Vocational Competency Certificate — RFID on Delivery Robot | SMK Telkom Jakarta | Score: 90/100 | January 2024

LANGUAGES

Indonesian — Native **English** — Intermediate | TOEIC Score: 415/990 | ETS, September 2023